

MAT1503

May/June 2019

Linear Algebra

Duration 2 Hours

100 Marks

EXAMINERS

FIRST

SECOND

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Closed book examination

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This paper consists of 4 pages

THE USE OF A POCKET CALCULATOR IS NOT PERMITTED

Answer all the questions

There is a total of 100 marks

[TURN OVER]

QUESTION 1

- (a) Show that the system

$$\begin{aligned}x + y + 2z &= 9 \\2x + 4y - 3z &= 1 \\3x + 6y - 5z &= 0\end{aligned}$$

has $(x, y, z) = (1, 2, 3)$ as a solution (3)

- (b) Show that the solution of the system in 1(a) is the only one for that system (4)

- (c) Consider the system

$$\begin{aligned}x_1 + 2x_2 + 3x_3 &= 5 \\2x_1 + 5x_2 + 3x_3 &= 3 \\x_1 + 8x_3 &= 17\end{aligned}$$

- (i) Write the coefficient matrix
- A
- of the system (1)

- (ii) Show that
- A
- is invertible (6)

- (iii) Verify that

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \quad (6)$$

- (iv) Use this inverse
- A^{-1}
- , to solve the system (5)

[25]

QUESTION 2

- (a) Using Cramer's rule to solve the system in question 1(a) (12)

- (b) Suppose that
- $A^2 = A$
- and prove that
- $\det(A) = 0$
- or
- $\det(A) = 1$
- (5)

- (c) Two matrices
- A
- and
- B
- are called
- SIMILAR**
- if there exists an invertible matrix
- P
- such

that $A = PBP^{-1}$. If A and B are similar, then prove that $\det(A) = \det(B)$ (5)

- (d) Given the matrix

$$C = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{bmatrix}$$

with $k = 3$, show that $\det(kC) = k^n \det(C)$ where n is such that C is an $n \times n$ matrix (3)

[25]

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QUESTION 3

(a) Consider the vectors $\underline{u} = (2, -1, 1)$, $\underline{v} = (1, 1, 2)$. Then

(i) compute $\underline{u} \cdot \underline{v}$ (2)

(ii) determine the angle θ between $\underline{u}$ and $\underline{v}$ (3)

(b) Let $\underline{u} = (2, -1, 3)$, $\underline{a} = (1, 0, 0)$ and find

(i) the orthogonal projection of $\underline{u}$ on $\underline{a}$ (4)

(ii) the vector component of $\underline{u}$ orthogonal to $\underline{a}$ (2)

(c) Find the distance between the two points $(1, 0, 0)$ and $(0, 1, 0)$ (3)

(d) Show that the relations $L_1: x = 1 + t, y = 3 - 2t, z = 1 - 4t$ and $L_2: x = 2 - 2t, y = 1 + 4t, z = -3 + 8t$ describe the same line (6)

(e) Suppose V_1 and V_2 are planes in $\mathbb{R}^3$ defined by

$$\begin{aligned} V_1: 2x + 3y + z &= 6 & \text{and} \\ V_2: 4x + 6y + 2z &= 9 \end{aligned}$$

Show that V_1 and V_2 are not the same plane (5)

[25]

QUESTION 4

(a) State De Moivre's Theorem (3)

(b) Let $z = \cos \theta + i \sin \theta$ be the polar form of the complex number z and n a natural number

(i) Show that $\frac{1}{z^n} = \cos n\theta - i \sin n\theta$ (4)

(ii) Show that $\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right)$ and (4)

(iii) Using (ii) show that $\cos^n \theta = \frac{1}{2^n} \left(z + \frac{1}{z} \right)^n$ (2)

[TURN OVER]

(iv) Use (iii) to show that

$$\cos^5 \theta = \frac{1}{2^5} [2 \cos 5\theta + 10 \cos 3\theta + 20 \cos \theta] \quad (12)$$

[25]

TOTAL: 100 Marks

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